

# Mohammed Huzaifa

## IoT & Embedded Systems Engineer

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### SUMMARY

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IoT engineer comfortable on both sides of the wire: embedded firmware and sensor hardware on one end, full-stack dashboards and protocol simulation on the other. Prefers building a whole prototype rather than one layer of it.

### SKILLS

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|---------------------------------|-------------------------------------------------------------------------------------------------------|
| <b>Embedded &amp; Hardware</b>  | Arduino, ESP32, GPIO, PWM, Interrupts, Timers, I2C, SPI, UART, PCB basics, power & battery management |
| <b>Wireless &amp; Protocols</b> | Wi-Fi, Bluetooth/BLE, LoRa, LoRaWAN, NB-IoT, MQTT, CoAP, TCP/IP                                       |
| <b>Languages &amp; Tools</b>    | Python, C/C++, Java, Git/GitHub, PlatformIO, Edge Impulse, AWS IoT, Azure IoT Hub                     |

### PROJECTS

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#### PULSE: Wearable Mood-Responsive Environment Control

*ESP32 · MQTT · Node.js · React/TypeScript*

- Built an ESP32 wearable that combines a GSR sensor (skin conductance), a MAX30102 (heart rate and SpO2), and an MPU6050 (6-axis IMU) over I2C and ADC to work out the wearer's arousal state in real time. The IMU is what lets it tell a real mood spike apart from someone just moving around.
- Wired up an MQTT to Node.js to WebSocket pipeline so sensor readings reach a React dashboard fast enough to drive lighting, music, and climate changes as the mood state updates.
- Checked the classification logic against a 30-session Monte Carlo simulation and wrote up every equation in the pipeline in an accompanying IEEE-format paper.

#### NetSim: Browser-Based LoRaWAN Network Simulator

*React · TypeScript · Web Workers*

- Simulated up to 50 battery-powered LoRaWAN devices talking to 3 gateways, with EU868 duty-cycle limits, Adaptive Data Rate (SF7 to SF12), and Semtech's time-on-air math modeled for real, not just for show.
- Modeled radio collisions with a capture effect, plus per-radio-state (sleep, RX, TX) power draw, so battery levels drop the way they actually would on real hardware.
- Ran the whole simulation inside a Web Worker and built a custom UI on top showing live topology, airtime budgets, and battery drain.

#### Smart Home Hub: Automation Rules Engine with 3D Simulation

*React · TypeScript · React Flow · React Three Fiber*

- Built a drag-and-drop rules engine (trigger, condition, and action nodes) that drives a live 3D floorplan, so devices react to automations the moment a rule fires.
- Added a connectivity layer with signal strength, battery drain, and random disconnects, wired into the rules engine itself, so a rule can't fire off a device that's stale or unreachable.
- Built a simulated command-security layer (per-device tokens, schema validation) with a working replay-attack demo to show how those protections actually fail.

### EDUCATION & TRAINING

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#### Samsung Innovation Campus, IoT Track, Presidency University, Bangalore

Selected through a competitive entrance test (27 of 350 applicants). Coursework: Fundamentals of IoT & Embedded Circuit Systems; Version Control & Collaborative IoT Software Development.

### EXTRAS

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- Smart India Hackathon: built a software attendance-automation system, placed in the university's top 20 of about 150 competing teams.
- Supplementary coursework: Game Design (Macquarie University), AI & IoT in Precision Agriculture.
- Volunteer, Green Bengaluru afforestation drive (June 2026), part of a Guinness World Record attempt to plant 1.5 million native saplings.
- Instructables: [instructables.com/ashuzaifa](https://www.instructables.com/ashuzaifa) · ORCID: 0009-0006-6229-3699